

Use Case Template

ChargEV – Smart EV Charging Station Platform

1. Use Case Title	Geocode Location
2. Abbreviated Title	Geocode Location
3. Use Case Id	UC-09
4. Actors	Location Services API (Supporting)
5. Description	The backend converts a free-text address or the driver's device coordinates into a normalised location usable for a station search.
5.1. Pre Conditions	1. A raw location string or coordinate pair has been received from the frontend.
5.2. Task Sequence	1. Backend receives the location input from the driver's search request. 2. Backend calls the Google Maps Geocoding API with the input. 3. API returns normalised latitude/longitude coordinates. 4. Backend uses the coordinates to drive the nearby-station query.
5.3. Post Conditions	1. Normalised coordinates are available for the station search that follows.
6. Modification History	Date: September 18, 2026
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